

1) Computer vision enables computers to capture, process and analyze videos to identify objects, recognize patterns, and make intelligent decisions using image processing and AI

a) Main objectives of computer vision

- Image understanding
- object detection and recognition
- scene analysis

b) Two Real-World applications

- Face Recognition
- Medical image analysis

② Computer vision extracts useful information from images using AI algorithms and AI techniques

a) Define feature extraction

feature extraction is a process of identifying important characteristics such as edges, corners, textures, and shape of an image.

b) object recognition enables computers to identify & classify objects accurately, improving automation and decision-making.

③ a) Low-level image processing involves basic operations such as image enhancement, noise removal, filtering.

b) Mid Level	High Level
Image Segmentation	Object Recognition.
Feature Extraction	Scene Understanding
Pattern detection.	Decision making

- ④ A digital image is represented as a matrix of pixels arranged in rows and columns.
- a) A pixel (picture element) is the smallest unit of a digital image that stores brightness or colour information.
 - b) Higher resolution provides more images details & better quality, while lower resolution reduces clarity.
- ⑤ Light reflected from an object passes through a camera lens onto an image sensor, which converts it into electrical signals.
- a) An image sensor converts light into electrical signals.
 - b) Digital camera, Scanner.
- ⑥
- a) Sampling converts a continuous image into discrete pixels.
 - b) Quantization converts continuous intensity values into discrete digital levels for storage & processing.
- ⑦ The image formation model explains how light from a real-world object is focused through a camera lens onto an image sensor to produce digital image.
- a) Perspective projection is the process of projecting a three-dimensional scene two-dimensional image plane.
 - b) Illumination provides light that enables objects to be visible and captured by camera.
- ⑧
- a) Image processing is the technique of enhancing, restoring, and manipulating digital images.

b) Two differences

Image processing	Computer vision
Improves Image quality	Understands image content
output is an enhanced image	output is information or decision.

9 a) Computer vision helps detect diseases, analyze medical images, and assist doctors in diagnosis.

b) Computer vision detect lanes, traffic signs, pedestrians, and vehicles to enable safe navigation.

10 Digital Image fundamentals provide the basic for Representing, processing, and analyzing images used in Computer vision.

a) pixel stores brightness and color information that collectively form a complete digital image.

b). Sampling determines the number of pixels in an image.

• Higher resolution provides more details, improving object detection and image analysis.

• Lower resolution reduces image quality and may affect recognition accuracy.

